

Entrega5: Application

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Librerías usadas

- TKinter, nos permite hacer la interfaz gráfica general
- Matplotlib, nos permite realizar los gráficos necesarios

```

# Pass all to a new table
for i in range(len(output)):

    # Look if the user was previously append
    userbool = False
    for j in range(len(dataframe)):
        if (dataframe[j][1] == output[i][1]):
            userbool = True

    if userbool == False:
        # IP(s)
        IPs = [output[i][2]]
        # Route(s) list
        RTs = [output[i][3]]

        # Append to the new table all data from the old one
        # Bool / User / IP(s) / Route(s) / Connection(s) / #IP(s) / #Route(s)
        input = [output[i][0], output[i][1], IPs, RTs, 0, 1, 1]
        dataframe.append(input)

    for j in range(len(output)):
        # Same user
        last = len(dataframe) - 1
        if (dataframe[last][1] == output[j][1]):
            # New Connection
            dataframe[last][4] += 1
            # If the IP associate is new append it in new table
            conf = False
            for k in range(len(dataframe[last][2])):
                if (dataframe[last][2][k] == output[j][2]):
                    conf = True

            if conf == False:
                dataframe[last][2].append(output[j][2])
                dataframe[last][5] += 1

            conf = False
            for k in range(len(dataframe[last][3])):
                if (dataframe[last][3][k] == output[j][3]):
                    conf = True

            if conf == False:
                dataframe[last][3].append(output[j][3])
                dataframe[last][6] += 1

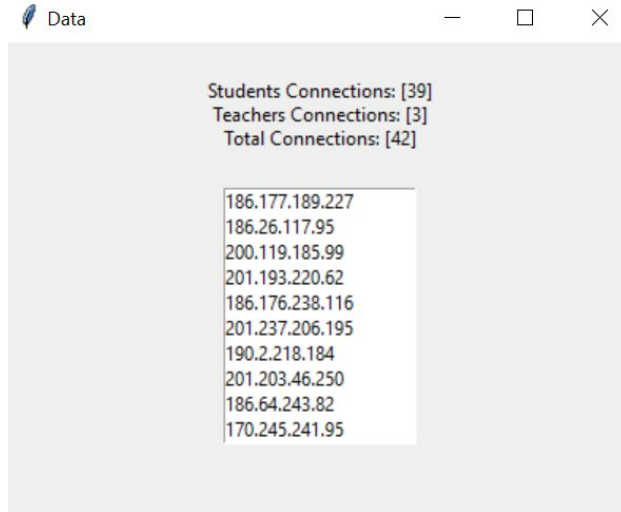
```

Se limpia la tabla base

```

['B70873', ['186.177.189.227'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-ne
['B62804', ['186.26.117.95'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net-
['b57617', ['200.119.185.99'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net
['B16802', ['201.193.220.62'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net
['B67751', ['186.176.238.116'], ['PUSH_REPLY,route::vpn-net::,route::net::255.255.0.0,rou
['anom.pelik', ['201.237.206.195'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::pri
['B66310', ['190.2.218.184'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::net::255.
['b78720', ['201.203.46.250'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net
['B77676', ['186.64.243.82'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net-
['b51558', ['170.245.241.95'], ['PUSH_REPLY,route::vpn-net::,route::net::255.255.0.0,rou
['B75612', ['186.176.25.38'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net-
['b60210', ['186.159.209.129'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-ne
['B62230', ['186.15.189.176'], ['PUSH_REPLY,route::vpn-net::,route::net::255.255.0.0,rou
['B83714', ['201.206.146.212'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-ne
['B70376', ['201.203.6.207'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::net::255.
['B60686', ['186.15.16.38'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::priv-net-3
['B71677', ['186.32.218.55'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::net::255.
['anom.hba', ['201.198.177.175'], ['PUSH_REPLY,route::vpn-net::255.255.255.0,route::net::
['b64390', ['186.4.62.56'], ['PUSH_REPLY,route::vpn-net::,route::net::255.255.0.0,route::

```



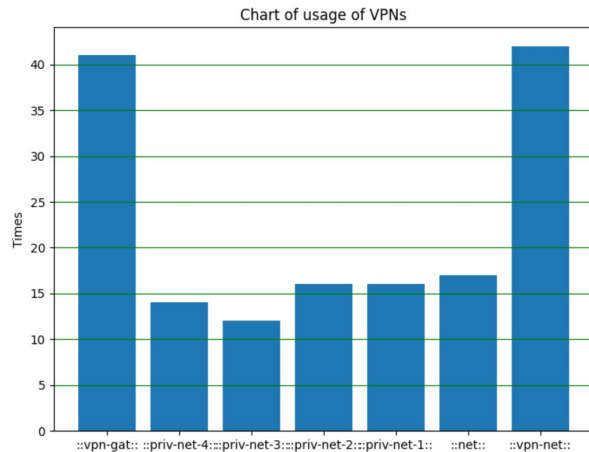
1. Conexiones en total

2. Lista de Direcciones
IP

3. Cantidad de estudiantes y profesores

4. Conexiones por cada ruta establecida

Figure 1



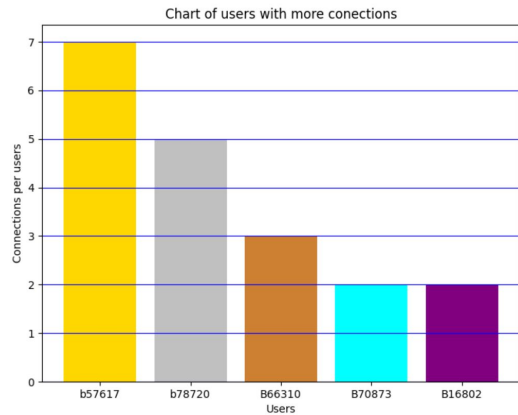
```
for item in vpns:
    new_vpn = item.replace("[", "").replace("]", "") # get rid of the [] symbols of the
    input = [item, 0]
    connection_list.append(input)

for connection in connection_list:
    benchmark = connection
    counter = 0
    for line in analyzer_conn:
        for vpn in line:
            if benchmark[0] == vpn:
                counter += 1
    benchmark[1] = counter

print (connection_list)
connection_vpns = [line[0] for line in connection_list]
vpn_amount = [line[1] for line in connection_list]
```

5. 5 usuarios con más conexiones

Figure 2



```
# 5. Los 5 usuarios con más conexiones realizadas (mostrar un gráfico).
```

```
# TODO: put this function in the new application code
```

```
# vector with the names
```

```
names = [line[1] for line in dataframe]
```

```
# vector with the connections( treated as ints)
```

```
conexions = [int(conexion[4]) for conexion in dataframe]
```

```
# sort the names and conexions
```

```
all_sort = sorted(list(zip(names,conexions)), key=lambda x: x[1], reverse=True)
```

```
nombres_ordenados , cantidades = zip(*all_sort)
```

```
# reduce the amount to the best 5
```

```
name_top5 = nombres_ordenados[:5]
```

```
amount_top5 = cantidades[:5]
```

Muchas Gracias